
FAMI-FIGHTER TURBO BOX

ファミファイター・ターボボックス



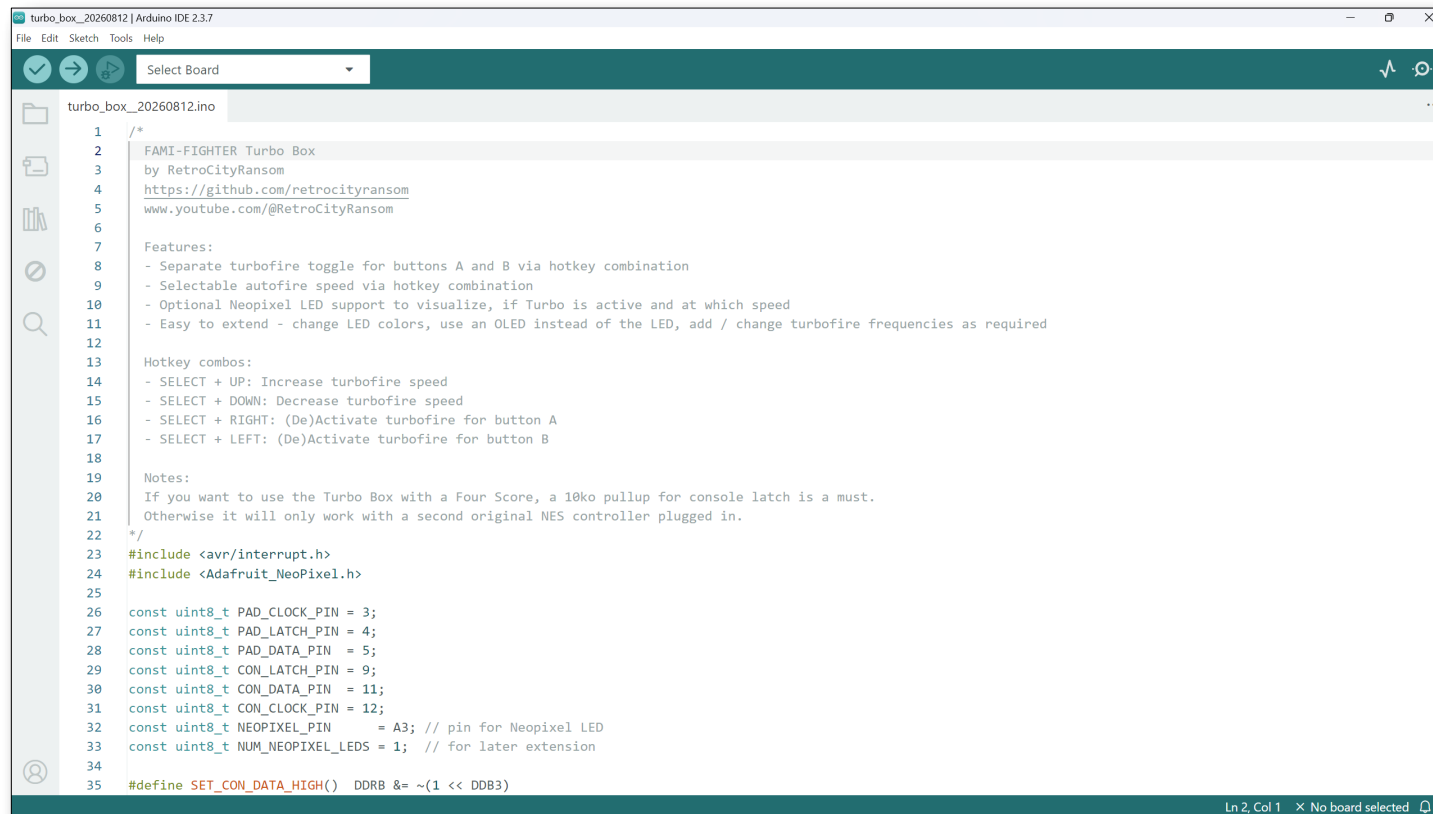
Flashing Instructions for Arduino
ver. 20260818

Prerequisites

- Download the code from <https://github.com/retrocityransom/fami-fighter>.
- Download and install the Arduino IDE from <https://github.com/arduino/arduino-ide/releases/>. It is free.
- Get a USB-to-TTL adapter for the Arduino Pro Mini. It must support 5 V operation. The examples in this guide use an IDUINO FT232.
- Important: It is best to flash the code to the Arduino before installing it in the Turbo Box. However, you can flash updates or changes afterwards without any issues. Just make sure to disconnect your Turbo Box from your game console before connecting it to the USB-to-TTL adapter.

Step 1

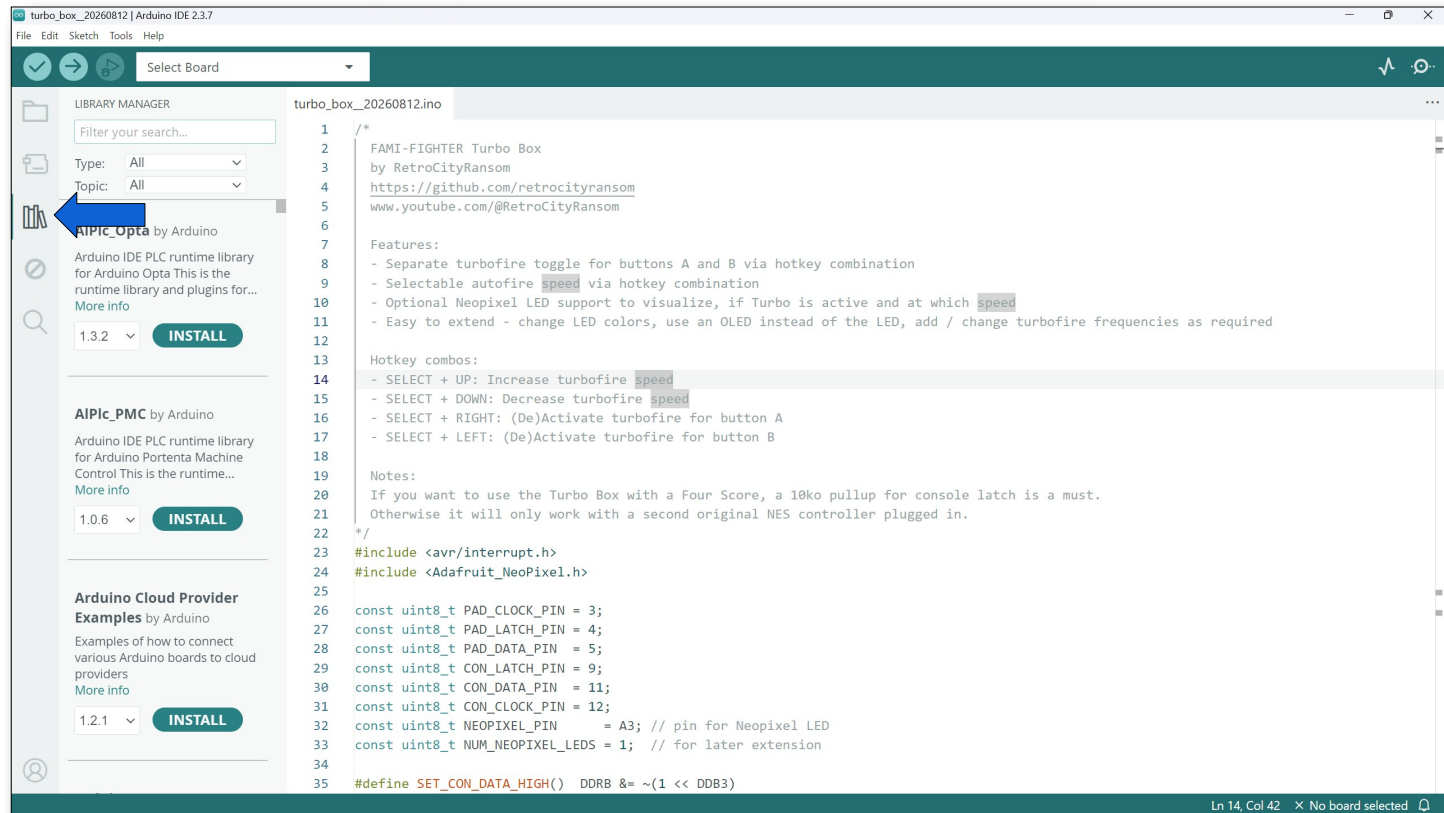
Open the .ino script ("sketch") from the code section that corresponds to the chosen hardware implementation variant. Currently, there is only one variant, but I plan to add a second one with an additional IC. Do not connect your USB-to-TTL adapter to your PC yet.



```
turbo_box_20260812.ino
1  /*
2   FAMI-FIGHTER Turbo Box
3   by RetroCityRansom
4   https://github.com/retrocityransom
5   www.youtube.com/@RetroCityRansom
6
7   Features:
8   - Separate turbofire toggle for buttons A and B via hotkey combination
9   - Selectable autofire speed via hotkey combination
10  - Optional Neopixel LED support to visualize, if Turbo is active and at which speed
11  - Easy to extend - change LED colors, use an OLED instead of the LED, add / change turbofire frequencies as required
12
13  Hotkey combos:
14  - SELECT + UP: Increase turbofire speed
15  - SELECT + DOWN: Decrease turbofire speed
16  - SELECT + RIGHT: (De)Activate turbofire for button A
17  - SELECT + LEFT: (De)Activate turbofire for button B
18
19  Notes:
20  If you want to use the Turbo Box with a Four Score, a 10ko pullup for console latch is a must.
21  Otherwise it will only work with a second original NES controller plugged in.
22  */
23  #include <avr/interrupt.h>
24  #include <Adafruit_NeoPixel.h>
25
26  const uint8_t PAD_CLOCK_PIN = 3;
27  const uint8_t PAD_LATCH_PIN = 4;
28  const uint8_t PAD_DATA_PIN = 5;
29  const uint8_t CON_LATCH_PIN = 9;
30  const uint8_t CON_DATA_PIN = 11;
31  const uint8_t CON_CLOCK_PIN = 12;
32  const uint8_t NEOPIXEL_PIN = A3; // pin for Neopixel LED
33  const uint8_t NUM_NEOPIXEL_LEDS = 1; // for later extension
34
35  #define SET_CON_DATA_HIGH() DDRB &= ~(1 << DDB3)
```

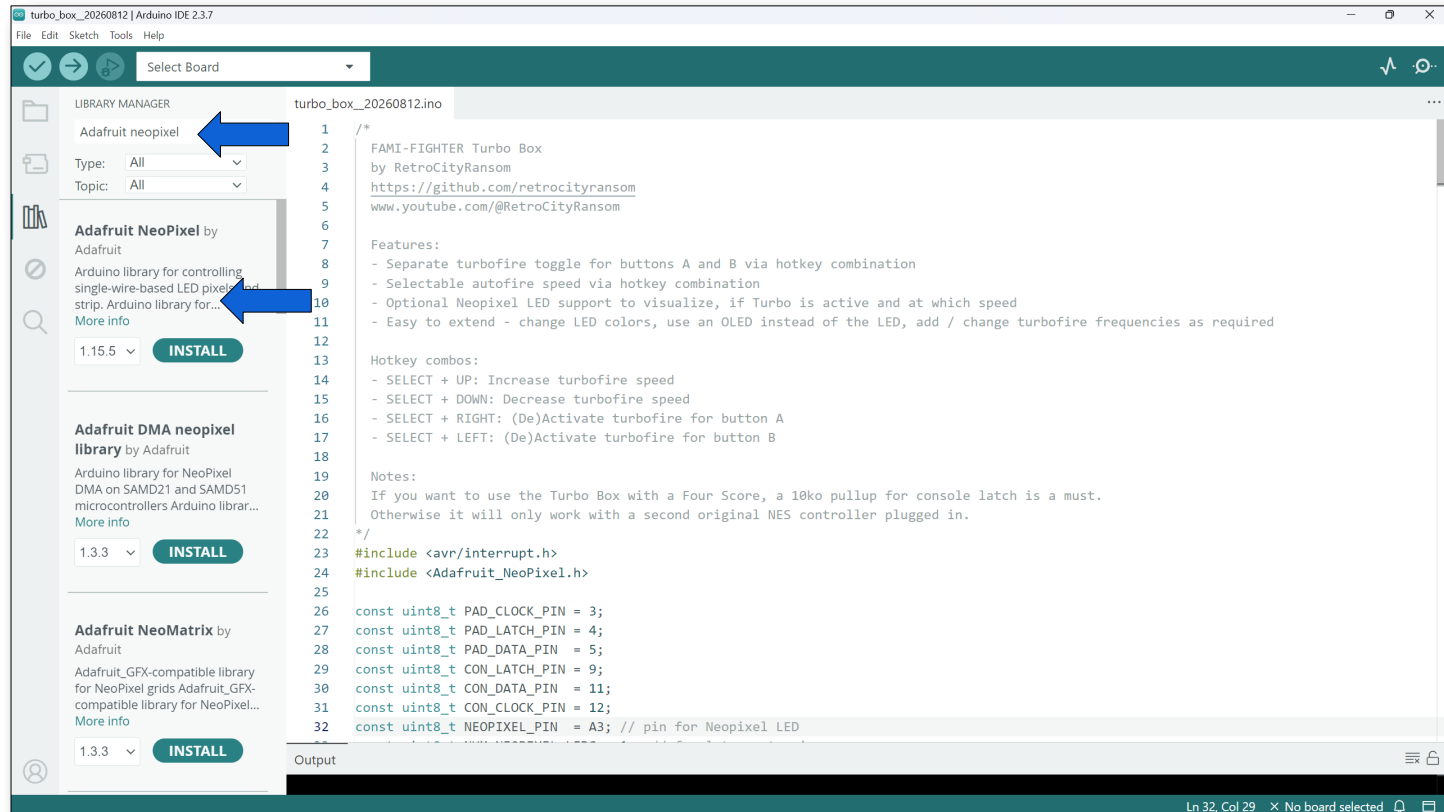
Step 2

To install the required libraries, click the Library icon on the left-hand side.



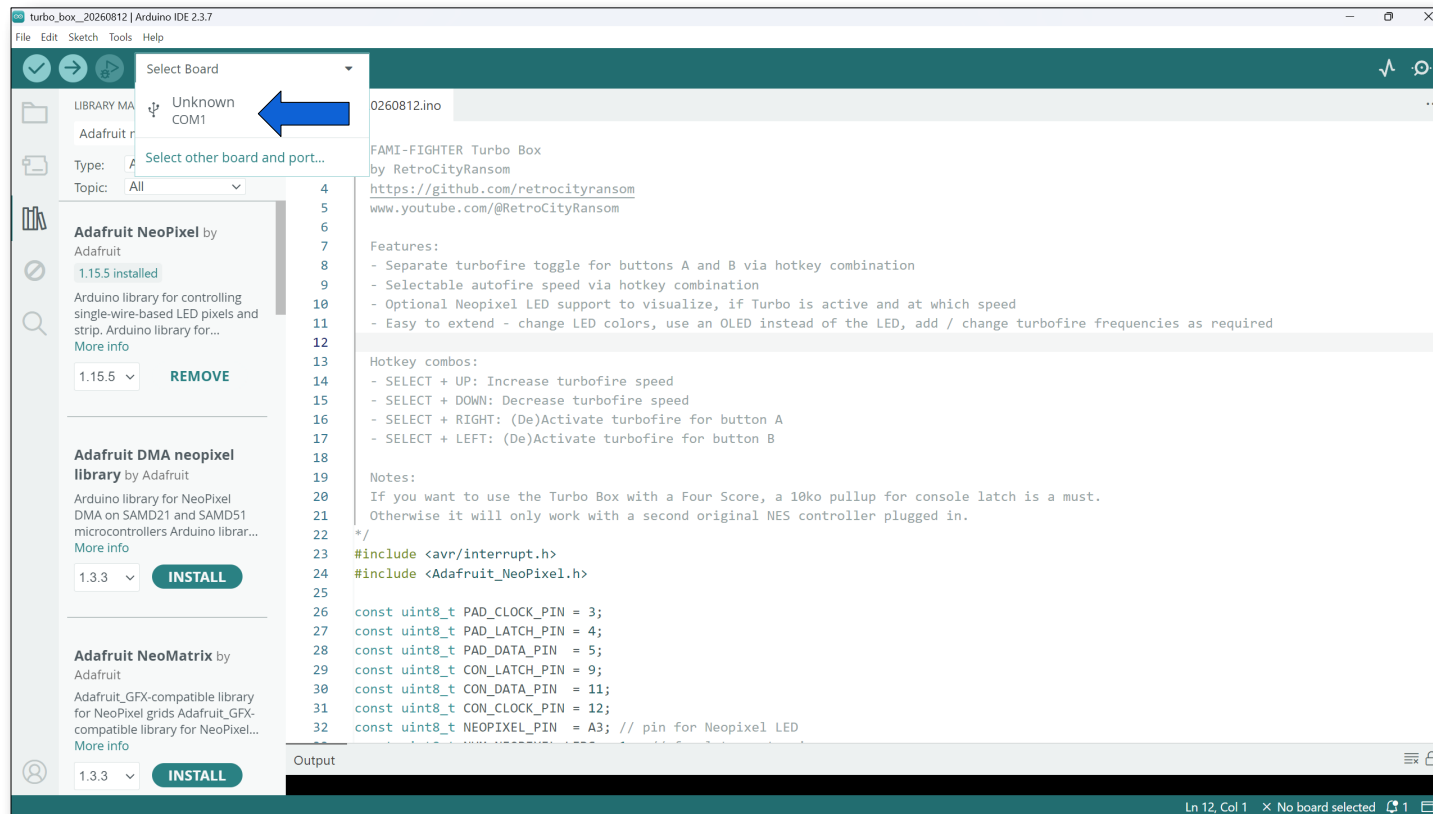
Step 3

Search for and install the Adafruit NeoPixel library.



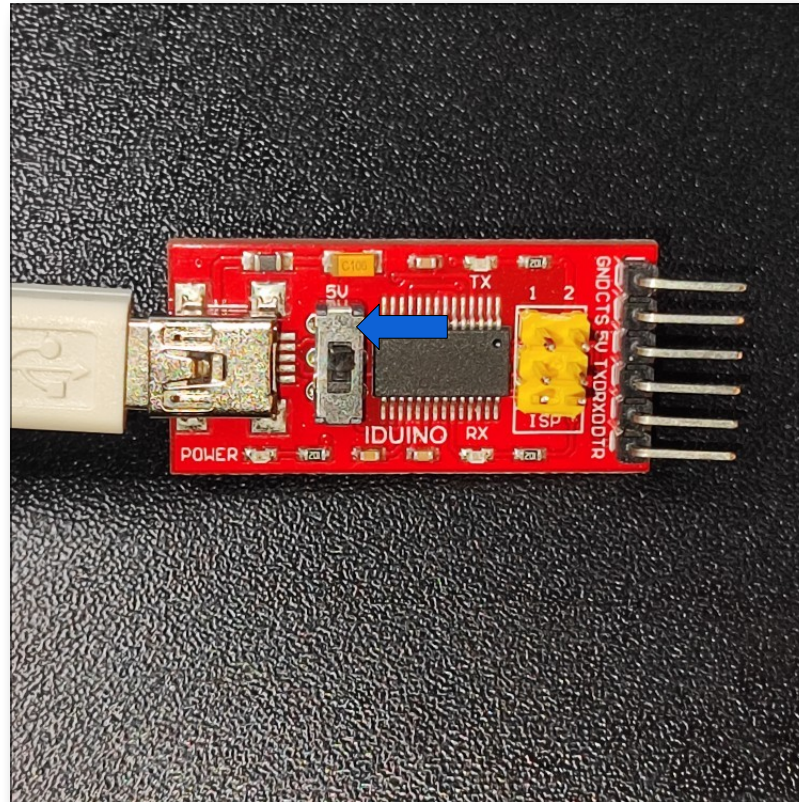
Step 4

With the USB-to-TTL adapter still disconnected from your PC, click on "Select Board" and note the COM port(s) shown. Once you connect your USB-to-TTL adapter via USB, an additional port should appear here.



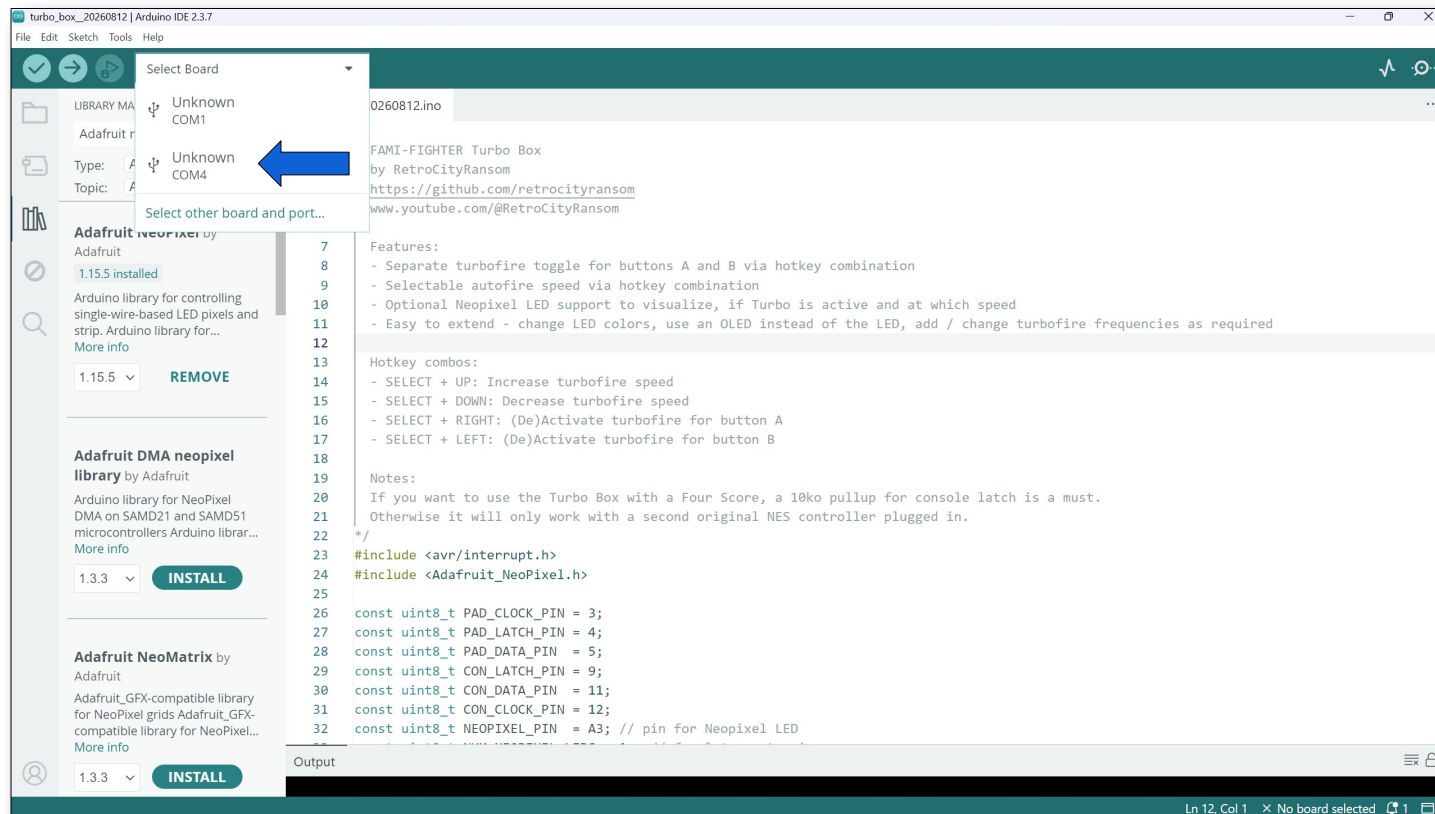
Step 5

Set your USB-to-TTL adapter to 5 V and connect it to your PC.



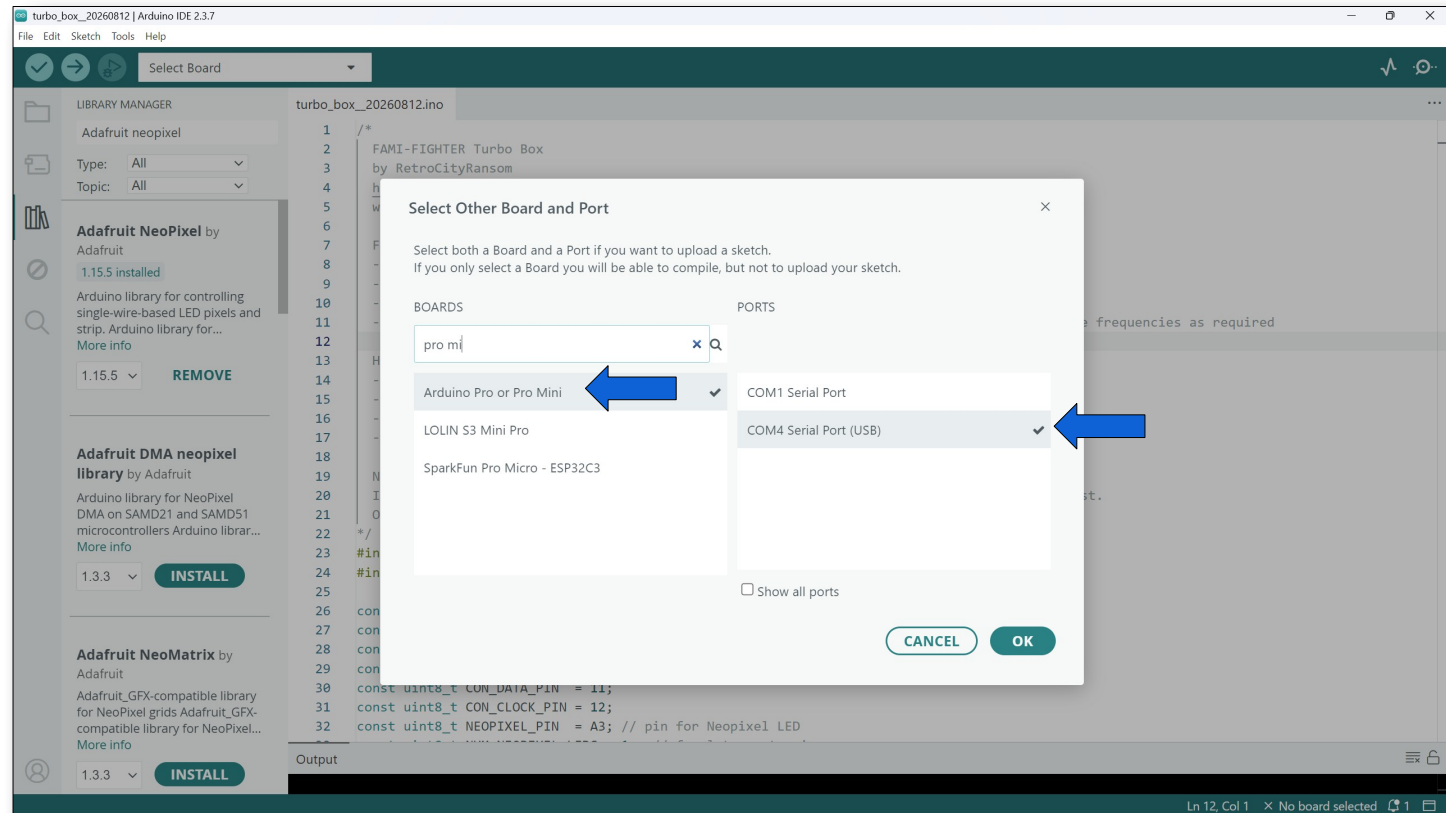
Step 6

Now select the new port that appeared after connecting your USB-to-TTL adapter. In this example, it is COM4, but it may be different on your PC.



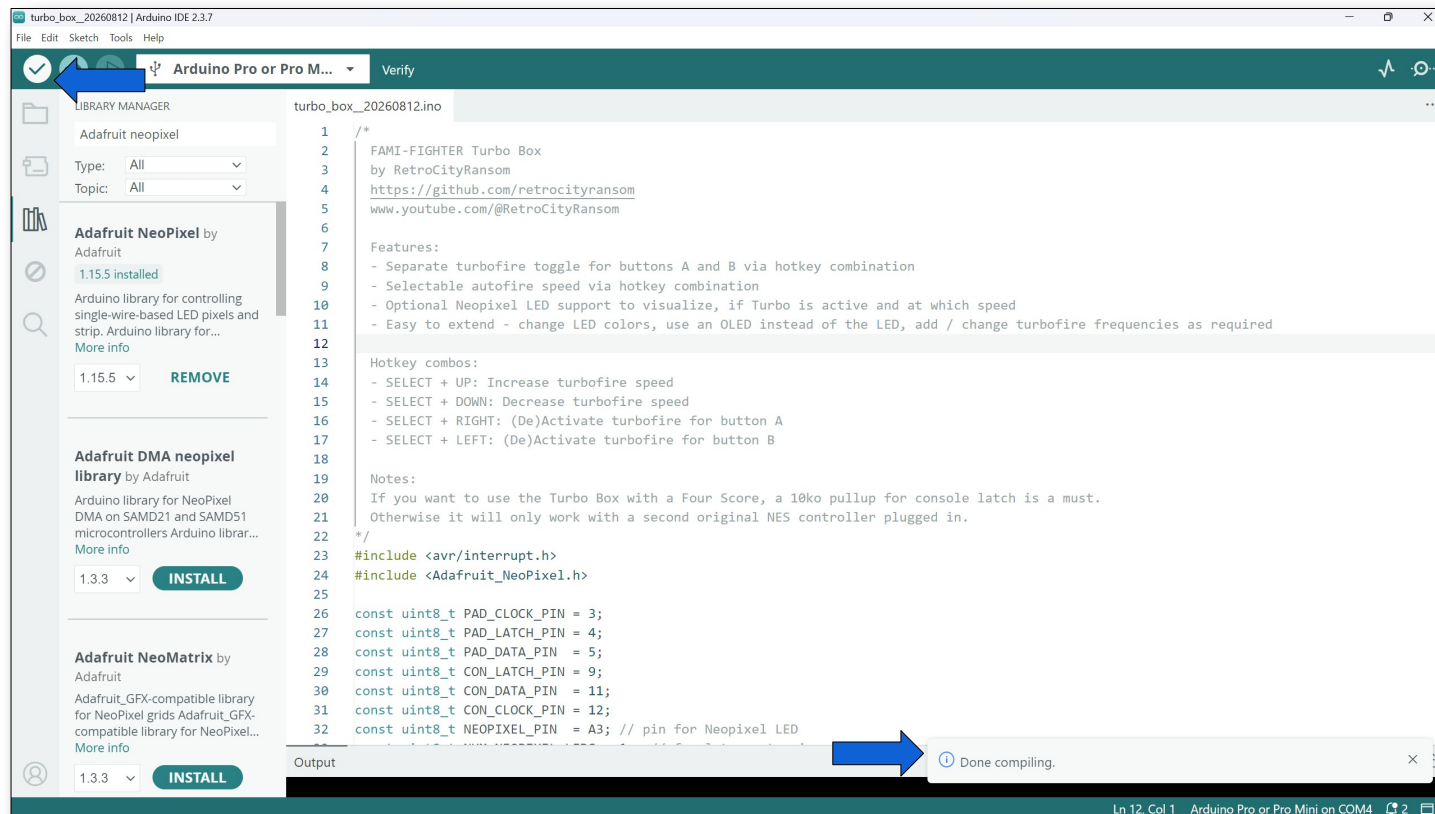
Step 7

A new dialog for selecting your board will appear. Select the Arduino Pro Mini option and make sure that the port assigned to your USB-to-TTL adapter is correct.

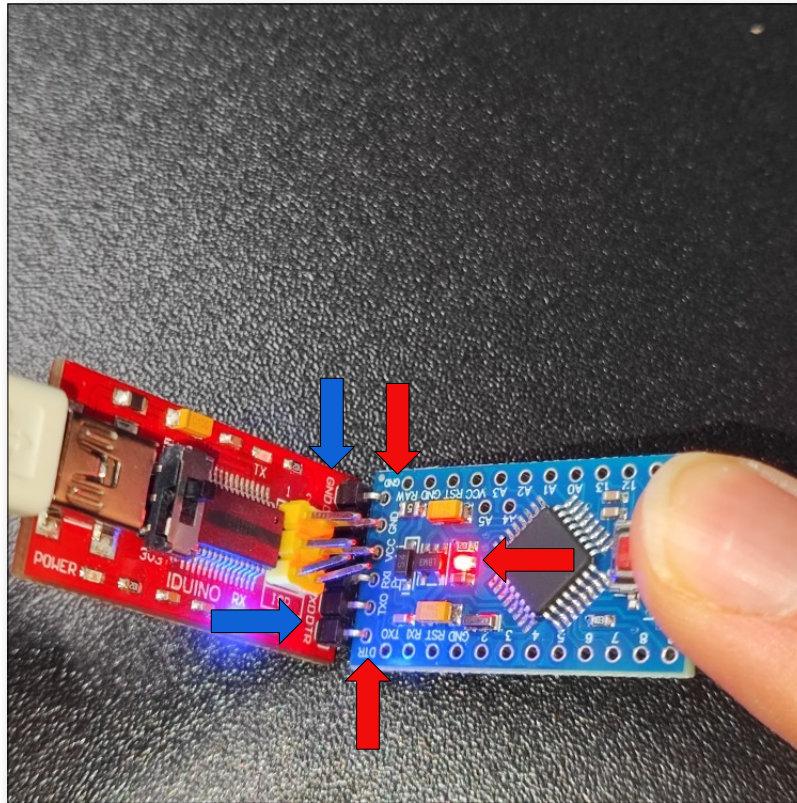


Step 8

You can now optionally verify the code. Click the checkmark icon in the top-left corner. If the code is syntactically correct, all required libraries have been installed, and the code is compatible with your selected board, you will see a message saying "Done compiling." in the bottom-right corner.



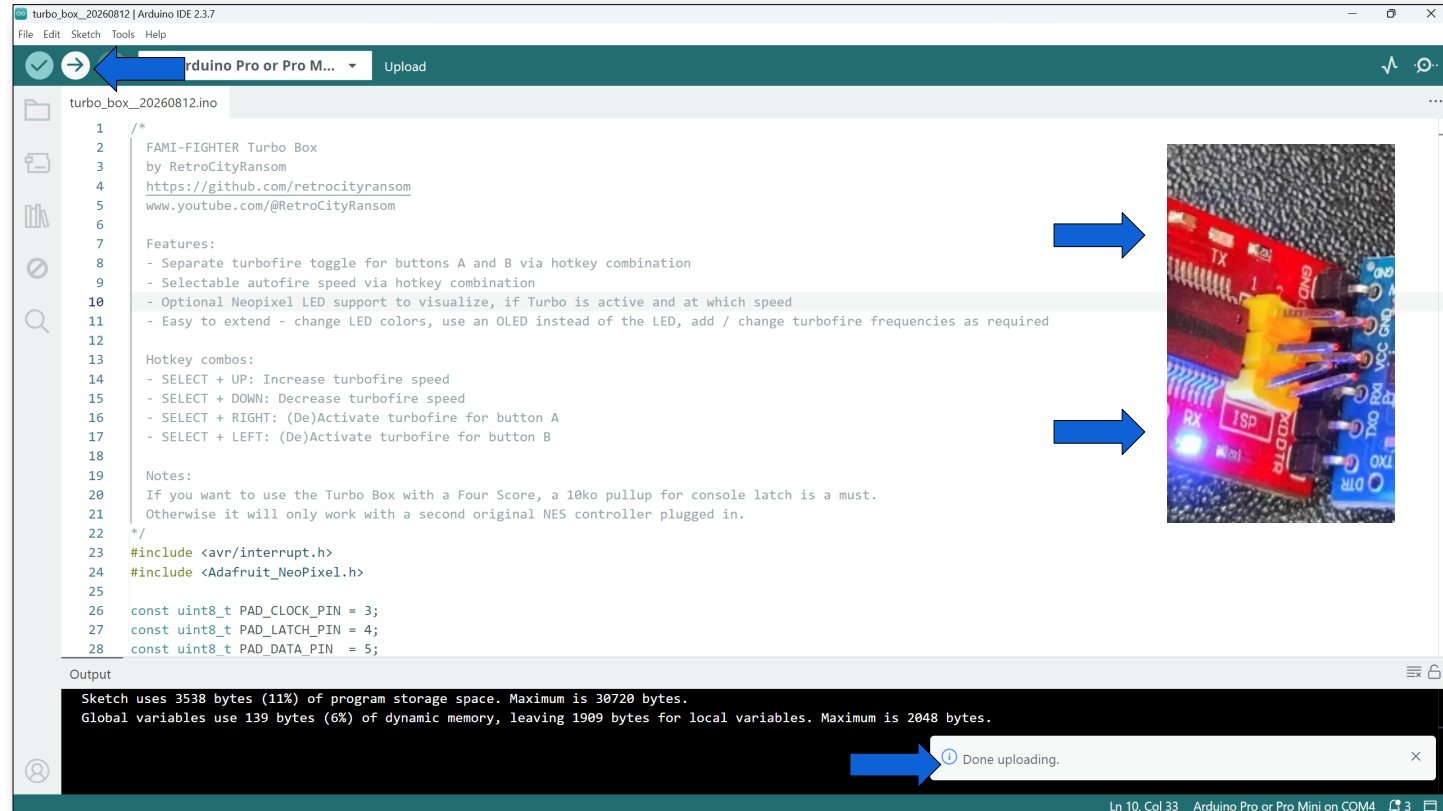
Step 9



Now it is time to connect your USB-to-TTL adapter to the Arduino and flash the code. Make sure that you connect the GND and DTR pins of the Arduino to their corresponding pins on the adapter. The red LED in the middle of the Arduino board should light up. If it does not light up, or if another LED on the Arduino board starts flashing, try gently applying a little pressure between the two boards to ensure that the pins make proper contact.

Step 10

Click the arrow icon to flash the code to the Arduino board. If your USB-to-TTL adapter is an IDUINO FT232, as shown in the example, two blue LEDs should start flashing during the programming process. The IDE will display a small notification in the bottom-right corner when the process has been completed successfully.



```
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28 const uint8_t PAD_DATA_PIN = 5;
```

Sketch uses 3538 bytes (11%) of program storage space. Maximum is 30720 bytes.
Global variables use 139 bytes (6%) of dynamic memory, leaving 1909 bytes for local variables. Maximum is 2048 bytes.

Done uploading.

Ln 10, Col 33 Arduino Pro or Pro Mini on COM4 3

Your Arduino is now ready to go.

Have lots of fun with it!

- **Code & diagrams:** <https://github.com/retrocityransom/fami-fighter>
- **Trailer:** <https://youtu.be/9fGX9Tb3N7Q>
- **Arduino IDE:** <https://github.com/arduino/arduino-ide/releases/>